Software Requirements Specification

Evidence-Based AI Performance Management Platform

Consolidated Documentation

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Confidential

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1 Scope

1.1 In-Scope

- **Data Integration & Schema Mapping:**
  - Automated schema extraction from custom enterprise databases (e.g., Claims, CRM, HRIS)
  - AI-driven schema mapping to standard KPI formats
  - Automatic generation of SQL data transformation queries
- **Automated OKR Tracking:**
  - Hierarchical OKR cascading (Company → HOD → Manager → Employee)
  - Real-time, event-based tracking of Key Results from ingested operational data
  - KPI engine calculations
- **Continuous Feedback & Workflow Context:**
  - Flexible employee check-in cadences (Weekly, Monthly, Quarterly)
  - AI-generated check-in summaries based on tracked work
  - Manual KPI overrides and qualitative context additions by employees
- **Performance Reviews & Analytics:**
  - Automated generation of quarterly performance summaries based on empirical baseline data
  - Eradication of recency bias through comprehensive long-term data analysis
  - Dashboard visualizations of team metrics, OKR completion rates, and organizational insights

1.2 Out-of-Scope

- Core HR payroll processing or compensation adjustments
- Recruiting, applicant tracking (ATS), or talent acquisition workflows
- Direct training modules or Learning Management System (LMS) hosting
- Direct modification or write-backs to the enterprise source databases (Read-only integration focus)
- General purpose enterprise chat platforms (acts instead as a performance overlay)

2 Functional Requirements

2.1 FR-1: Universal Data Ingestion Engine

2.1.1 FR-1.1: Schema Extraction

- System SHALL connect to enterprise databases using secure connectors (e.g., PostgreSQL, Snowflake, API connections).
- Schema Extraction Agent SHALL query database metadata (e.g., `information_schema`) to read the structural format of the tables.
- System SHALL output schema metadata in standardized JSON formats.

2.1.2 FR-1.2: Intelligent Schema Mapping

- Schema Mapping Agent SHALL utilize an LLM to compare custom enterprise columns with the platform's unified `employee_activity` schema.
- SQL Transformation Agent SHALL generate valid SQL translation queries to convert source events into target KPIs.
- The Data Pipeline SHALL automatically ingest the transformed data at scheduled intervals.

2.2 FR-2: Strategic OKR Engine

2.2.1 FR-2.1: Hierarchical Alignment

- System SHALL allow Executive users to create top-level Organizational Objectives.
- System SHALL enforce a downward cascade, allowing HODs, Managers, and Employees to link personal Key Results to parent Objectives.

2.2.2 FR-2.2: Automated Event Tracking

- The OKR Engine SHALL recalculate progress continuously based on the incoming stream of structured KPI activities.
- Employees SHALL NOT be required to manually calculate or submit numerical progress on automated metrics.

2.3 FR-3: Contextual Check-in System

2.3.1 FR-3.1: Configurable Check-in Frequency

- Employees SHALL be able to select checking frequency configurations: Weekly, Monthly, or Quarterly.
- The AI Insights Agent SHALL generate a unified textual summary of accomplishments aligned with that period's ingested data.

2.3.2 FR-3.2: Human-in-the-Loop Overrides

- During check-in validation, Employees SHALL be able to modify the drafted response to add supplementary context.
- Employees SHALL have the ability to report "Manual KPIs" representing non-digitized efforts directly into the check-in form.

2.4 FR-4: AI-Generated Performance Summaries

2.4.1 FR-4.1: Empirical Review Generation

- At the conclusion of a performance cycle, the AI Review Agent SHALL evaluate all ingested data, calculated KPIs, and stored check-in contexts.
- System SHALL generate structured performance documents identifying Strengths, Areas of Improvement, and Evidence-backed ratings.
- Line managers SHALL review, edit, and approve these AI-drafted reviews prior to delivery.

3 System Architecture

3.1 Architectural Layer Overview

The platform operates as a multi-layered Intelligence Ecosystem.

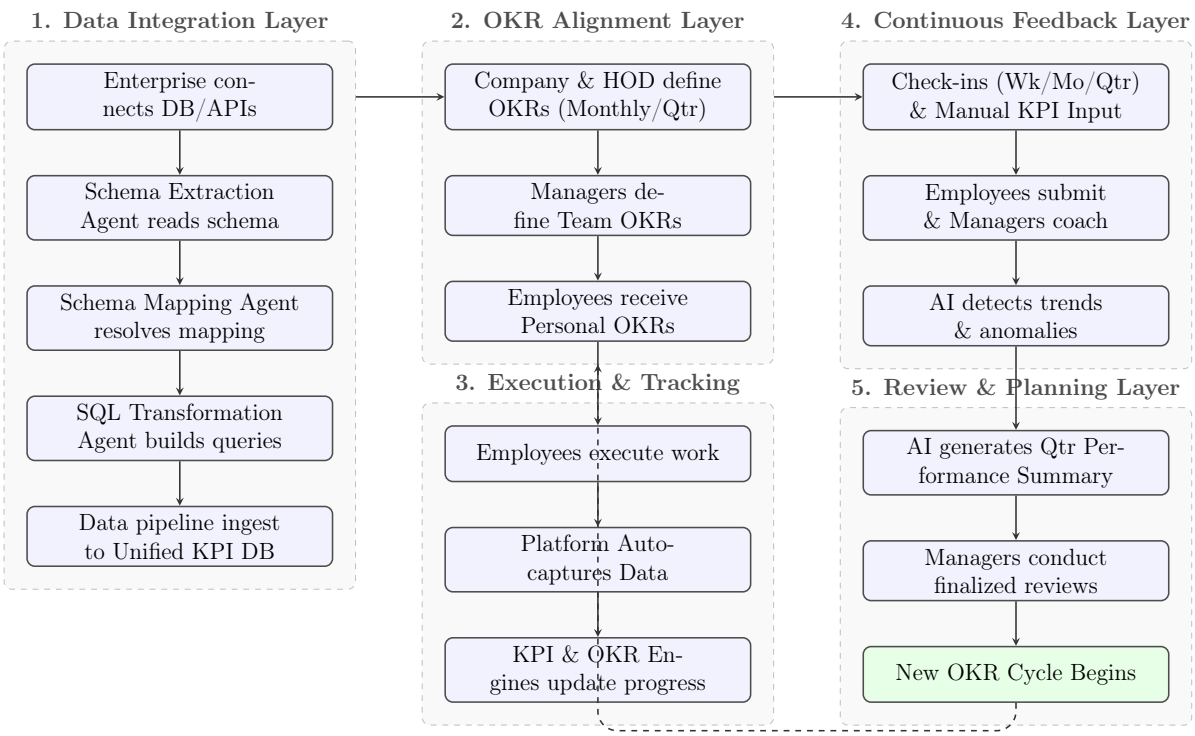

Figure 1: End-to-End Artificial Intelligence Performance Management Ecosystem Workflow

1. **Data Integration Layer:** AI Agents (Extraction, Mapping, Transformation) process raw operational business data into unified Key Performance Indicators.
2. **OKR Alignment Layer:** Business logic linking Company → Department → Team → Individual priorities.
3. **Work Execution & Tracking Layer:** Passive data collection and real-time metric updates.
4. **Continuous Feedback Layer:** Configurable check-in loops heavily augmented by conversational AI.
5. **Review & Planning Layer:** Executive analytics dashboards and quarter-end automated review cycles.

3.2 Detailed Agentic Workflows

3.2.1 Workflow 1: The Automated Schema Mapping Flow

1. **Trigger:** Enterprise connects a new data source (e.g., proprietary Claims Oracle DB).
2. **Extraction:** Schema Extraction Agent reads `table_name`, `column_name`, `data_type`.
3. **Mapping:** Schema Mapping Agent identifies logical keys (e.g., `agent_id` → `employee_id`).
4. **Execution:** SQL Transformation Agent generates an ELT script. Data begins syncing automatically without manual engineering bandwidth.

3.2.2 Workflow 2: The Continuous Feedback Loop

1. **Aggregation:** When a check-in cadence is reached (e.g., end of the Month), the AI aggregates all operational actions the employee completed.
2. **Drafting:** AI drafts a summary: "You resolved 145 tickets and achieved 80% of Key Result 1."
3. **Human Context:** Employee reviews the draft, inputs context (e.g., "Was delayed by vendor outages"), and submits.
4. **Coaching:** Manager reviews the enriched data and provides targeted coaching feedback.

4 Competitive Analysis & Market Landscape

4.1 Market Landscape

The traditional performance management software market is highly fragmented, split primarily between lightweight continuous feedback tools (JOP, Lattice) and heavy Strategy Execution platforms (Peoplebox).

4.2 Competitor Deep Dive: Peoplebox

Focus: Strategy & OKR Execution.
Peoplebox attempts to solve alignment problems by mapping company strategy down to employee performance. It relies heavily on standard API integrations (Jira, Salesforce) to drive OKR updates, while maintaining traditional quarter-end manager reviews.

Weakness: It relies entirely on standard REST APIs, making it impossible to integrate with the archaic, custom legacy databases of major enterprises (e.g., an insurance company's 15-year-old on-premise Claims system). Despite integrations, final reviews are often subjective HR-driven exercises.

4.3 Competitor Deep Dive: JOP (Joy of Performing)

Focus: Continuous Performance Culture.
JOP dismantles the annual review entirely, pushing teams toward high-frequency (weekly) 1:1 check-ins, 360-degree peer feedback, and soft-skill competency tracking.

Weakness: Almost entirely devoid of hard data. Check-ins rely on employees staring at a blank page and self-reporting their week's progress. Because it relies heavily on subjective manager feedback and peer agreement, it does not eradicate human biases.

4.4 Competitive Comparison Matrix

Feature	**Our AI Agent Platform**	**Peoplebox**	**JOP**
Data Ingestion	**Autonomous Schema Agents**	Standard API Config	Manual Entry
Execution Baseline	**100% Objective (Hard Data)**	Mixed (Data + HR process)	Subjective Feedback
Check-in Format	**AI-Drafted + Manual Context**	Manager-Driven 1:1s	Manual Self-Reporting
Performance Bias	**Eradicated via historical data**	High (Manager calibration)	High (Peer calibration)
Custom DB Support	**Yes (AI-mapped SQL Generation)**	No (Only standard SaaS)	No

5 Innovation Strategy

5.1 Differentiation from the Status Quo

To dominate the enterprise market—especially highly regulated, data-heavy industries like Insurance or Finance—we pivot away from "culture surveys" and focus entirely on **Evidence-Based Intelligence**.

1. **Schema-Agnostic Integration:** The ability to plug immediately into highly customized enterprise databases removes months of implementation friction and makes the product instantly valuable to CTOs.

2. **Eradicating the Blank Page Problem:** The most widely reported failure of tools like JOP is that employees hate writing weekly check-ins. By having the AI

pre-draft the check-in based on the actual work done, we achieve 100% adoption with minimal friction.

3. **Objective, Legally Defensible Reviews:** By basing quarterly performance algorithms on hard operational data points mapped to OKRs, organizations avoid the toxicity of "recency bias" or subjective popularity contests often created by 360-degree feedback tools.

5.2 The "Agentic" Shift

The core innovation is moving Performance Management from a **Passive HR Reporting Tool** to an **Active Analytical Co-Pilot**.

- Rather than asking employees to report on their goals, the Agents *tell the employees* how they are performing against their goals.
- Rather than managers struggling to remember 3 months of an employee's work, the Review Agent synthesizes the exact evidentiary timeline of productivity.